

Helmershus archive

AdvanReader-10 via Python (Mercury API)

What we actually did to talk to Keonn AdvanReader-10 + Advantenna-p16 from Python on Linux. Official Keonn SDKs are Java / C# / C++. Python uses the community wrapper gotthardp/python-mercuryapi, compiled against the Novanta/ThingMagic Mercury API C SDK.

Hardware

- AdvanReader-10 (USB, FTDI, RS232-over-USB) + Advantenna-p16
- Linux device: /dev/ttyUSB0 (kernel FTDI driver, no extra driver)
- User must be in dialout: `sudo usermod -aG dialout $USER` then log out/in
- URI: `eapi:///dev/ttyUSB0` baud: 115200 region: EU3
- Weak USB ports brown out / timeout under RF - use a strong port (~1 A)

Linux install (the path that worked)

No virtualenv. The mercury module was installed into system Python.

1. Clone the helmershus repo and `cd python-mercuryapi-master`
2. Drop the Novanta SDK zip next to the Makefile: `mercuryapi-BILBO-1.37.2.24.zip`
(Makefile curl URL is dead - download manually)
3. Install build deps:

```
sudo apt-get install -y python3 python3-dev python3-setuptools \
    build-essential unzip patch xsltproc libreadline-dev gcc
```

4. Build and install against that zip (do not pip install from PyPI):

```
make APIZIP=mercuryapi-BILBO-1.37.2.24.zip APIVER=1.37.2.24
sudo make install APIZIP=mercuryapi-BILBO-1.37.2.24.zip APIVER=1.37.2.24
python3 -c "import mercury; print('ok')"
```

Build fixes we hit

- Missing xsltproc - LTKC header generation failed (xsltproc: No such file or directory)
- SDK 1.37 vs wrapper: in mercury.c Reader_write_tag_mem, TMR_writeTagMemBytes needs a 7th argument. Add NULL:

```
ret = TMR_writeTagMemBytes(&self->reader, tag_filter, bank, address,
    PyByteArray_Size(data), (uint8_t *)PyByteArray_AsString(data), NULL);
```

- License classifier / PyEval_InitThreads messages were warnings only
- mercuryapi.patch on 1.37 applied with fuzz; that was OK

Repo

- Wrapper + SDK build tree stayed local on Linux
- Root .gitignore: python-mercuryapi-master/
- sudo make install is outside git; import mercury survives pulling
- When first gitignoring the folder, move it aside before pull if mercury.c was edited locally

Runtime

- advanreader_simple.py: connect, set_region("EU3"), set_read_plan([1], "GEN2", read_power=...), read(500) loop
- Power is centi-dBm (1500 = 15 dBm). Hardware max ~2700; USB current often limits it
- Check range: reader.get_power_range()
- Java Keonn examples (eapi:///dev/ttyUSB0 + RXTX natives) also worked; Python Mercury was enough once USB power was OK

Do not repeat

- Mac compile (LTKC / stdint: _STDINT_H vs _STDINT_H_)
- pip install python-mercuryapi (dead JADAK zip URL)
- tmr://IP - this unit is USB serial, not Ethernet
- Leaving Java open on the same /dev/ttyUSB0 while Python connects

Archived 21 Aug 2026. UHF Keonn path is no longer used; Helmershus continues with two Nexmosphere stations.